# Efficient Beltrami Flow in Patch-Space

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## Contribution

- Extension of Beltrami filter into patch-space.
- Efficient scale-space traversal of the image manifold.
- Demonstration of a simple empirical relationship between the image intensities and their coordinates as posed in [1].

## Problem statement

The Beltrami framework treats images as two dimensional manifolds embedded in a hybrid spacial-spectral five dimensional $\{x, y, R, G, B\}$ space. Denoising the image can then be effectively performed through mean curvature area minimization of this embedded surface.

How can we extend this approach so as to treat local image pixels as part of the feature space?

## Beltrami framework

**Functional**

Minimizing the functional

$$S[X, G] = \iint \sqrt{g} dx dy$$

leads to an Euler-Lagrange equation which can be compactly written as

$$-div\left(\sqrt{g} G^{-1} \nabla I\right) = 0.$$

Freedom of parameterization permits appending $g^{-1/2}$ which yields the second order differential operator of Beltrami

$$\Delta_g I = 0.$$

**Metric**

Equating distances on the manifolds produces the induced metric tensor

$$G = \begin{pmatrix} 1 + I_x^2 & I_x I_y \\ I_x I_y & 1 + I_y^2 \end{pmatrix}.$$

## Patch-space metric

**Patch definition**

We define a patch of size w×w as

$$P(x, y) = \left(x, y, \left\{I^k(x + iw, y + jw)\right\}\right)$$

where $i, j = -w, .., w,\ k = 1, .., n$.

**New patch metric**

Using this extended feature space we can derive the new induced metric tensor as

$$G = \begin{pmatrix} 1 + \beta^2 \sum_{i,j,k} {I^k_{i,j}}^2 x & \beta^2 \sum_{i,j,k} I^k_{i,j} x I^k_{i,j} y \\ \beta^2 \sum_{i,j,k} I^k_{i,j} x I^k_{i,j} y & 1 + \beta^2 \sum_{i,j,k} {I^k_{i,j}}^2 y \end{pmatrix}$$

**Image flow**

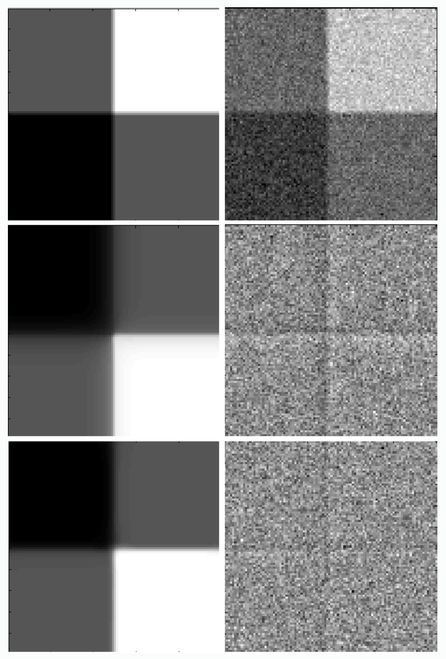
The image can now be denoised by the flow

$$I_t = \Delta_g I = \frac{1}{\sqrt{g}} \det\left(\sqrt{g} G^{-1} \nabla I\right).$$

## Results



Examples of Beltrami patch denoising for color images. From top to bottom, left to right a) Noisy F16, $\sigma = 20$ b) Denoised image, $PSNR = 31.51dB$ c) Noisy Lena, $\sigma = 30$ d) Denoised image, $PSNR = 29.54dB$ e) Noisy Mandrill, $\sigma = 50$ f) Denoised image, $PSNR = 21.43dB$



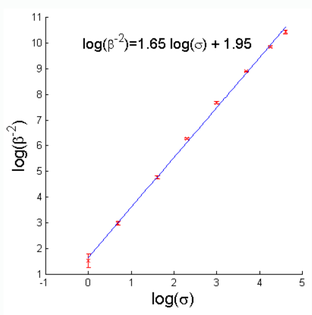
Method noise. a) Original image b) Noisy, $\sigma = 20$ c) Regular Beltrami denoising d) residual e) Beltrami patch denoising f) residual.

| | CMan | Lena | Barbara | House | |
|---|---|---|---|---|---|
| Regular Beltrami | 36.97 | 36.90 | 35.85 | 36.92 | $\sigma = 5$ |
| Patch Beltrami | 37.70 | 38.11 | 37.01 | 38.16 | |
| BM3D | 38.29 | 38.72 | 38.31 | 39.83 | |
| Regular Beltrami | 27.22 | 29.03 | 26.20 | 28.98 | $\sigma = 20$ |
| Patch Beltrami | 29.35 | 32.05 | 28.71 | 32.13 | |
| BM3D | 30.48 | 33.05 | 31.78 | 33.77 | |
| Regular Beltrami | 20.55 | 23.00 | 20.95 | 22.63 | $\sigma = 50$ |
| Patch Beltrami | 23.83 | 27.13 | 23.52 | 26.88 | |
| BM3D | 26.12 | 29.05 | 27.23 | 29.69 | |

PSNR comparison between Regular Beltrami, Patch Beltrami and BM3D for some standard grayscale images. All values are in dB

Patch Beltrami produces denoised images which are visually plausible. Furthermore the noise reduction achieved is generally within 2dB of the BM3D[2] method for reasonable values of $\sigma$. In all cases tested the method significantly improves on the results produced by Regular Beltrami denoising.

## Log-linear relationship




A simple log-linear relationship between $\sigma$ and $\beta^{-2}$ was determined empirically.

## Run times

| | $N = 256$ | $N = 512$ |
|---|---|---|
| $K = 1$ | 0.14 | 0.54 |
| $K = 5$ | 0.65 | 2.60 |
| $K = 10$ | 1.27 | 5.13 |
| $K = 20$ | 2.55 | 10.37 |
| $K = 50$ | 6.42 | 25.45 |



Our iPhone 4 implementation benefits from the $O(KN^2)$ run time complexity achieved by using an integral image.

## Aliasing reduction



a) Photograph of truck with aliasing. b) CBM3D denoising c) Beltrami Patch denoising d) Castle with weak block textures e) BM3D denoising f) Beltrami Patch denoising

The method presented tends to align colors along boundaries and is seen to reduce aliasing in images. The flow also traverses a scale space which flattens out smooth, weakly textured structures. It should be noted that a state of the art denoising method, BM3D, copes very poorly with these two situations because it is optimized for Gaussian noise removal alone.

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## References

[1] N. Sochen, R. Kimmel and R. Malladi A general framework for low level vision, In *IEEE Trans. on Image Processing '98*

[2] K. Dabov,A. Foi,V. Katkovnik and K. Egiazarian Image denoising by sparse 3-d transform-domain collaborative filtering, In *IEEE Trans. on Image Processing '07*